

Stage 5 - Trend Decomposition

trend_analysis.mlx

Fits a 3-year moving average to one crime category from the South African crime statistics dataset for separation of underlying trend from year-to-year noise by:

- Loading the sa_crime_clean CSV from data/processed/sa_crime_clean.csv,
- Extracting the Contact crime incidents and aggregating it to a national annual series,
- Applying a 3-year centered moving average to smooth year-to-year variation in the Contact series,
- Producing a visualisation displaying the historical series, smoothed moving average, a linear trendline, and COVID-19 anomaly, and
- Exporting and saving the produced visualisation to matlab/ for the technical reporting stage in the project pipeline.

Data source: Crime Statistics of South Africa (2011-2023)

Starts each script execution on a clean state:

```
clear all %#ok<CLALL>
clc
```

Navigates from the live script to the project root, i.e. since the script lives in matlab/, goes up one level to reach project root:

```
active_script = matlab.desktop.editor.getActiveFilename;           % gets
full directory path of live script
script_dir = fileparts(active_script);                             % gets
'matlab/' directory by filename stripping
project_root = fileparts(script_dir);                              % goes
up one level to project root
cd(project_root)                                                    %
changes path to project root

disp(pwd)
```

```
C:\Users\Wits-User\Desktop\PROJECTS\sa-crime-statistics
```

Loads processed data from file into data table:

```
import_opts = detectImportOptions('data/processed/sa_crime_clean.csv');
import_opts.VariableNamingRule = 'preserve';                       %
preserves original column names

sa_crime_tbl = readtable('data/processed/sa_crime_clean.csv', import_opts);

disp('Crime statistics dataset read into table!')
```

```
Crime statistics dataset read into table!
```

Filters sa_crime_tbl to the crime category Contact and aggregates incident counts to a national annual series:

```
contact_idx = strcmp(sa_crime_tbl.crime_category, 'Contact');           % array
of indices with row entries being in 'Contact' category
contact_tbl = sa_crime_tbl(contact_idx, :);

crime_years = unique(contact_tbl.financial_year);                     %
expected: [2011, 2012, ..., 2021, 2022]

% creates function to aggregate crime counts by year for crime totals:
crime_totals = arrayfun(@(yr)
sum(contact_tbl.incident_count(contact_tbl.financial_year == yr)),
crime_years);    % passes 'crime_years' as input array

disp([crime_years, crime_totals])
```

2011	615935
2012	608724
2013	611574
2014	616973
2015	623223
2016	608321
2017	601366
2018	617210
2019	621282
2020	535217
2021	607163
2022	653659

Applies a 3-year centered moving average to crime_totals for year-to-year noise smoothing, and fits a linear trend to the smoothed data. The slope of the linear trend provides an indication of the average change in annual Contact crime incidents across the 12-year period:

```
sliding_window = 3;           % moving
window length

smoothed_contact = movmean(crime_totals, sliding_window);           % 3-year
centered moving average

linear_fit = polyfit(crime_years, crime_totals, 1);                 % fits
linear trend

trend_line = polyval(linear_fit, crime_years);                       % evaluates
trend yearly

fprintf('Trend slope: %.0f incidents per year.\n', linear_fit(1))    % displays
linear trend slope
```

Trend slope: -479 incidents per year.

Produces the trend decomposition visualisation in the form of a multi-line chart with the following components: the historical Contact crime series, the 3-year moving average, and the linear trendline, with the COVID lockdown period specifically annotated:

```

covid_yr = find(crime_years == 2020); % index for
COVID lockdown period
covid_count = crime_totals(covid_yr); % crime
total for COVID lockdown period

figure('Units', 'inches', 'Position', [0, 0, 9, 6]) % inches for cleaner look

plot(crime_years, crime_totals, '-o', 'Color', '#2E7D32', 'LineWidth', 2,
'MarkerFaceColor', '#2E7D32', 'DisplayName', 'Actual')
hold on
plot(crime_years, smoothed_contact, '-.o', 'Color', '#1565C0', 'LineWidth', 2,
'MarkerFaceColor', '#1565C0', 'DisplayName', '3-year Moving Average')
plot(crime_years, trend_line, ':', 'Color', '#595959', 'LineWidth', 2,
'DisplayName', 'Linear trend')

% adds COVID annotation:
annotation('textarrow', ...
[0.72, 0.82], ...
[0.30, 0.17], ...
'String', sprintf('COVID-19\nLockdown'), ...
'Color', '#DC143C', ...
'FontSize', 11, ...
'FontAngle', 'italic')

% chart formatting:
xlabel('Year', 'FontSize', 11)
ylabel('Total Crime Incidents (Thousands)', 'FontSize', 11)
title('Contact Crime - Trend Decomposition (2011-2022)', ...
'FontSize', 13, 'FontWeight', 'bold')

xticks(crime_years)

% formats y ticks to thousands annotation:
vals_yticks = get(gca, 'YTick');
yticklabels(arrayfun(@(y_vals) sprintf('%d', round(y_vals/1e3)), vals_yticks,
'UniformOutput', false)) % returns cell array

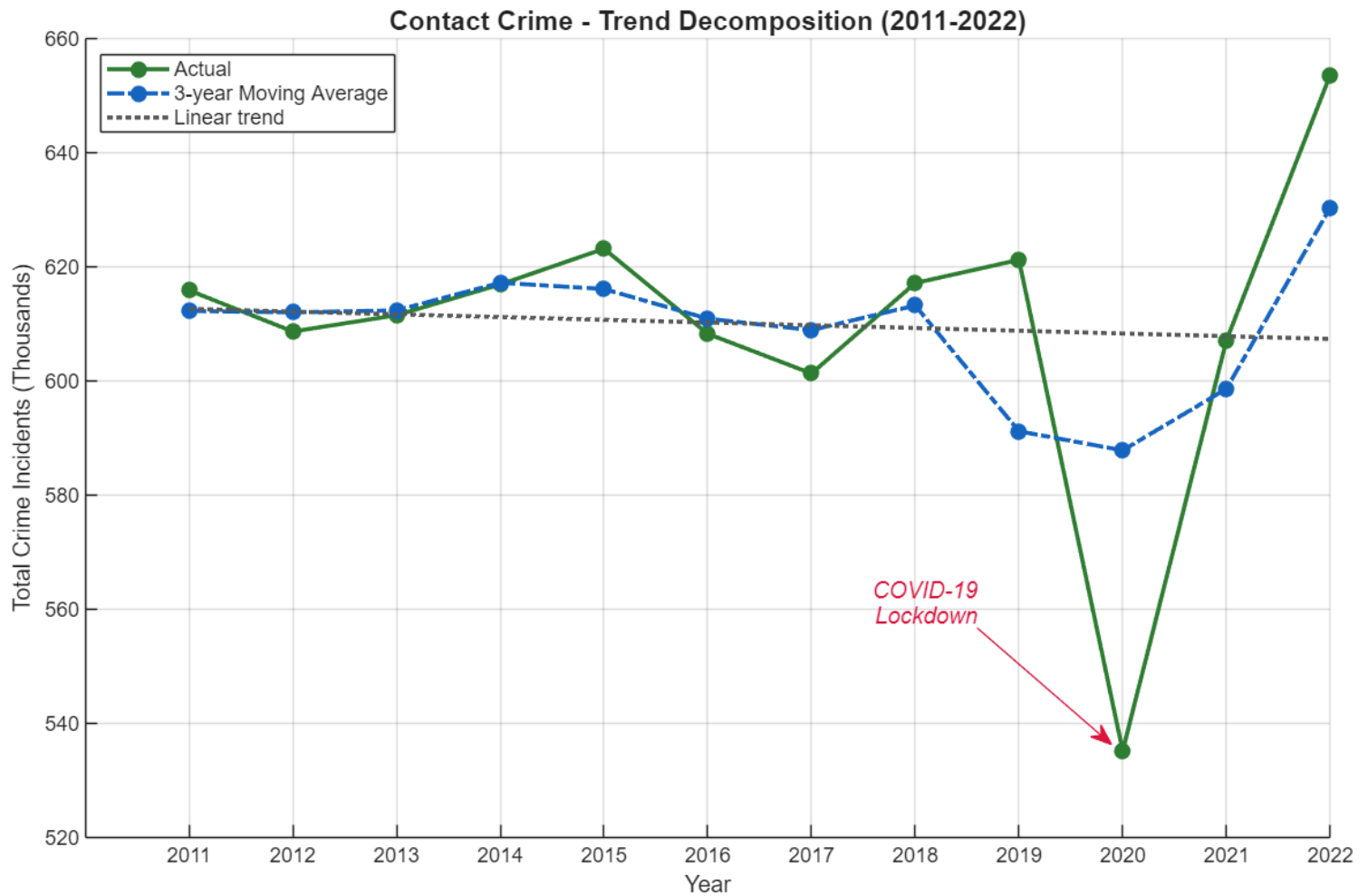
set(gca, 'LooseInset', get(gca, 'TightInset'))

legend('Location', 'northwest', 'FontSize', 10)
grid on
box off
hold off

```

Exports and saves produced chart to file:

```
print('-dpng', '-r300', 'matlab/trend_analysis.png')
print('-dpng', '-r300', 'reports/figures/trend_analysis.png')
```



```
disp('Export complete: matlab/trend_analysis.png')
```

Export complete: matlab/trend_analysis.png

The output of this script are as follows: one visualisation trend_analysis.png saved to matlab/. This singular output is ready for use in the technical reporting stage of the project pipeline.